



Pi

Draw a circle. Measure around it, then measure across. Divide the big number by the small one, and what do you get? The answer is 3 and a bit, or to be precise, pi. Humble pi, as it turns out, is one of the most remarkable numbers of all.



What is Pi?

Pi is simply the *circumference* of a circle divided by the *diameter*. It works out the same for all circles, no matter how big they are. Test this for yourself with a bit of string. Use the string to measure the distance around cups, buckets, plates, and so on, and divide the length of the string by the distance across.

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exactly

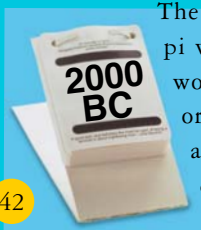
AN IRRATIONAL NUMBER

One of the weird things about pi is that you can't work it out exactly. There's no simple ratio, like $22 \div 7$, that equals pi exactly. That makes pi an "irrational" number. If you wrote it out in full (which is impossible), its decimal places would continue forever.



3.141592

The HUNT for Pi



The Egyptians reckoned pi was $16^2/9^2$, which works out as $256/81$, or 3.16. Not bad, but accurate to only one decimal place.



Greek philosopher Archimedes drew 96-sided shapes around circles and so worked out that pi is between $220/70$ and $223/71$ – accurate to 3 decimal places.



Ludolph van Ceulen worked out pi to 35 decimal places in Germany. But he died before the number was published, so it was carved on his grave.